Sharp-cube Class-II source-square reduction

TECT verification-first repository · claim A12-CLASSII-SOURCE-SQUARE-REDUCTION

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1 Purpose and scope

This note advances the first open gate left by the A11 true-increment determinant. On the three-torus of side length $L = 16$, write

$$S_j = P_{\le j}, \qquad u_j = S_{j-1}\phi, \qquad \ell_j = G_j^* B(u_j) D u_j, \qquad G_j = D P_j R_\Lambda C^{1/2}.$$

The filtration is the strict sharp rectangular-cube filtration

$$Q_j = \{n \in \mathbb{Z}^3 : \ |n|_\infty \le N_j\}, \qquad N_j = 2^j,$$

with disjoint shell projector $P_j = S_j - S_{j-1}$. The A11 first shell has $u_1 = \phi_0 = 0$; changing the harmless starting index only adds a finite low block.

The result proved here is the cutoff-uniform analytic estimate

$$\boxed{\sum_j \|G_j^* B(S_{j-1}\phi) D S_{j-1}\phi\|_2^2 \le C_{\mathrm{src}} \|\phi\|_6^6.} \tag{1.1}$$

The constant is explicit in two precisely defined sharp-cube L^6 operator norms. This note does not supply a certified decimal enclosure of those norms. Consequently it does not complete T-047, prove a positive production sextic reserve, or prove the A7 Nelson bound. The surviving quantitative gate is `A12-CLASSII-SHARP-CUBE-L6-VECTOR-NORM-ENCLOSURE`.

2 Pinned conventions

Let $\alpha_L = 2\pi/L$ and $k_n = \alpha_L n$. The A1 production symbol obeys

$$A(k) \ge c_{\mathrm{sym}}(1 + |k|^2)^2 I, \qquad c_{\mathrm{sym}} = 0.10836453058711468. \tag{2.1}$$

The scalar real-even regulator satisfies $\|R_\Lambda\|_\infty \le M_R$. The Class-II matrix acts on six real field components. Spatial derivative directions are included in the Hilbert norm; no extra factor three is inserted. The complex covariance factor already fixed by A7–A11 is not changed here.

For the production numbers set

$$a = \frac{c_{JJ}\alpha_X^2}{M_X^2 + \delta_X}, \quad b = \frac{c_{JK}\alpha_X\beta_X}{M_X^2 + \delta_X}, \quad c = \frac{c_{KK}\beta_X^2}{M_X^2 + \delta_X}. \tag{2.2}$$

The executable derives

$$a = 0.004499999999998875, \quad b = 0.0018749999999995311, \quad c = 0.002343749999999414. \tag{2.3}$$

3 The sharp Pauli/Fierz operator constant

Write $\psi = (z, w) \in \mathbb{C}^2 \oplus \mathbb{C}$, $s = |z|^2$, $\rho = s + |w|^2$, and $d = \rho + \varepsilon$. For a variation $h = (\eta, \omega)$ and the three Pauli matrices σ_A, define

$$m_A = z^* \sigma_A z, \qquad J_A = 2\,\mathrm{Re}\langle \sigma_A z, \eta\rangle, \qquad q_A = \frac{m_A}{d}, \qquad K_A = J_A - q_A\, d\rho. \tag{3.1}$$

Here $d\rho = ds + dt$, with $ds = 2\,\mathrm{Re}\langle z, \eta\rangle$ and $dt = 2\,\mathrm{Re}(\bar{w}\omega)$. Pauli completeness gives

$$|m|^2 = s^2, \qquad m \cdot J = s\, ds, \qquad |J|^2 \le 4s|\eta|^2. \tag{3.2}$$

An exact expansion, including the fixed floor, is

$$|K|^2 = |J|^2 - ds^2 + \left(ds - \frac{s}{d} d\rho\right)^2. \tag{3.3}$$

Put $a_0 = s/d$. Since

$$(1 - a_0)^2 + a_0^2 \frac{|w|^2}{s} = \frac{(|w|^2 + \varepsilon)^2 + s|w|^2}{d^2} \le 1, \tag{3.4}$$

Cauchy–Schwarz and $|dt| \le 2|w||\omega|$ show that the last square in (3.3) is at most $ds^2 + 4s|\omega|^2$. Thus

$$|J|^2 \le 4s|h|^2, \qquad |K|^2 \le 4s|h|^2, \qquad |J \cdot K| \le 4s|h|^2. \tag{3.5}$$

The production quadratic form is

$$h^T B(\psi) h = a|J|^2 + 2bJ \cdot K + c|K|^2.$$

Because $b > 0$, (3.5) yields the global operator inequality

$$\boxed{0 \le B(\psi) \le \beta_{\mathrm{op}} |\psi|^2 I_6, \qquad \beta_{\mathrm{op}} = 4(a + 2b + c) = 0.0423749999999894.} \tag{3.6}$$

It is sharp: take $w = 0$ and a doublet tangent direction with $d\rho = 0$ which saturates Pauli completeness. Then $K = J$ and equality holds. This is substantially sharper than the older Frobenius envelope $\beta_B = 0.2564999999999359$, while remaining fully compatible with it.

4 Exact sharp-shell decay

Shell j lies strictly outside the previous cube. Therefore every nonzero shell mode satisfies

$$|k_n| \ge \kappa_j := \alpha_L(N_{j-1} + 1). \tag{4.1}$$

The $N_{j-1} + 1$ is load-bearing: replacing it by an ambiguous continuous boundary loses the exact low-shell convention. From (2.1),

$$\begin{aligned} \|G_j^* F\|_2^2 &\le M_R^2 \sup_{n \in Q_j \setminus Q_{j-1}} \frac{|k_n|^2}{c_{\mathrm{sym}}(1 + |k_n|^2)^2} \|F\|_2^2 \\ &\le \frac{M_R^2}{c_{\mathrm{sym}}(1 + \kappa_j^2)} \|F\|_2^2. \end{aligned} \tag{4.2}$$

If a starting shell contains $n = 0$, D^* annihilates it and the remaining nonzero modes use $|n| \ge 1$. Equation (4.2) contains M_R^2: the source has one G_j, subsequently squared. A9's Hilbert–Schmidt term has two covariance legs and hence M_R^4; transferring that fourth power here is an error.

5 Sharp-cube vector-valued theorem

Define the dimensionless Hilbert-valued operator norms

$$M_6 := \sup_{f\neq 0} \frac{\| \sup_j |S_j f| \|_6}{\|f\|_6}, \tag{5.1}$$

and

$$Q_6 := \sup_{f\neq 0} \frac{\left\| \left(\sum_j \frac{|DS_{j-1}f|^2}{1+\kappa_j^2} \right)^{1/2} \right\|_6}{\|f\|_6}. \tag{5.2}$$

The Hilbert space includes six real field components and all three derivative directions. Both constants are finite and independent of the terminal cutoff.

For completeness, choose a product de la Vallée Poussin multiplier V_j which equals one on Q_j and vanishes outside a twice enlarged cube. The maximal V_j operator is controlled by iterated one-dimensional Fejér maximal operators. The remainder $R_j = S_j - V_j$ has uniformly finite cube-annular overlap. For arbitrary Rademacher signs, the multiplier

$$\sum_j \epsilon_j (1_{Q_j} - V_j) \tag{5.3}$$

has uniformly bounded mixed discrete variation on every signed dyadic rectangle. The periodic Hilbert-valued product Marcinkiewicz theorem and Khintchine's inequality therefore give

$$\left\| \left(\sum_j |R_j f|^2 \right)^{1/2} \right\|_6 \leq C\|f\|_6. \tag{5.4}$$

Since $\sup_j |R_j f|$ is bounded pointwise by the square function, (5.1) follows. This is a dyadic smooth-plus-annular argument; it does not assume a multiparameter Carleson theorem.

For (5.2), let $\Delta_k = S_k - S_{k-1}$. Random signed normalized shell multipliers

$$\sum_k \epsilon_k 2^{-k} \alpha_L^{-1} D\Delta_k \tag{5.5}$$

obey the same product Marcinkiewicz condition, including the cube faces. Moreover, for $k < j$,

$$\frac{\alpha_L 2^k}{\sqrt{1+\kappa_j^2}} \leq 2^{k-j+1}. \tag{5.6}$$

The scale kernel on the right has finite ℓ^1 norm. Vector-valued Littlewood–Paley plus scale Young proves (5.2). Sharp radial balls are not covered, and shell ratios tending to one would destroy this uniform scale summability.

There is an independent six-linear interpretation. After the pointwise operator bound (3.6), polarization reduces the scalar envelope to a finite sum of six-linear forms. With normalized geometric weights their multiplier is

$$m_J(n_1, \ldots, n_6) = \sum_{j\leq J} 2^{-2j}(n_5 \cdot n_6) \prod_{r=1}^{6} 1_{\{|n_r|_\infty \leq 2^j - 1\}}. \tag{5.7}$$

If $K = \max_r |n_r|_\infty$ and j_0 is the first admitted scale, the geometric sum and $|n_5 \cdot n_6| \leq 3K^2$ give $|m_J| \leq 1$. Uniform dyadic face variation then invokes the multilinear product Marcinkiewicz/paraproduct theorem. This second route is useful as an adversarial check, but the maximal/square route keeps the six-real tensor structure transparent. Neither route expands the rational B in Fourier series.

6 Source-square conclusion

Apply (4.2) and (3.6) with $u_j = S_{j-1}\phi$:

$$\begin{aligned}\sum_j \|\ell_j\|_2^2 &\le \frac{M_R^2\beta_{\rm op}^2}{c_{\rm sym}} \int \left(\sup_j |u_j|\right)^4 \sum_j \frac{|Du_j|^2}{1+\kappa_j^2} \\ &\le \frac{M_R^2\beta_{\rm op}^2}{c_{\rm sym}} M_6^4 Q_6^2 \|\phi\|_6^6.\end{aligned} \tag{6.1}$$

Thus (1.1) holds with

$$\boxed{C_{\rm src} = \frac{\beta_{\rm op}^2}{c_{\rm sym}} M_R^2 M_6^4 Q_6^2 = 0.016570372383568618\, M_R^2 M_6^4 Q_6^2.} \tag{6.2}$$

The rational fixed-floor coefficient is harmless here because only the exact pointwise inequality (3.6) is used. No polynomial band-limit shortcut occurs.

7 Production budget target, not closure

The A11 conditional determinant contributes at most $p^2 C_{\rm src}\|\phi\|_6^6/2$ from the source, while the production sextic contributes $p\gamma\|\phi\|_6^6/6$. Ignoring every other cost, source-only absorption requires

$$C_{\rm src} < \frac{\gamma}{3p}. \tag{7.1}$$

At $p = 1.1$, $M_R = 1$, and $\gamma = 1.62$, this becomes

$$M_6^4 Q_6^2 < 29.62571266025876. \tag{7.2}$$

This is a falsifiable numerical target, not an established inequality. If the obsolete A10 allocation $\epsilon_6 = 0.25$ were retained, the target would be 2.194497234093243. That allocation belongs to the refuted historical route and is not reused. The full budget must be recomputed only after a certified enclosure of $M_6^4 Q_6^2$.

8 Executable verification

The primary route derives a, b, c, $\beta_{\rm op}$, $c_{\rm sym}$, the exact shell boundary, M_R^2 scaling, rational non-bandlimiting, and the budget table directly from A1. It stress-tests the Fierz identities with the floor and third component active. The independent route imports neither the primary route nor A10 code; it rebuilds the Pauli metric, checks a finite six-linear multiplier family, and verifies the finite-cutoff Hölder step on a multiscale field. These numerical fixtures audit conventions. They do not replace the infinite-cutoff product Marcinkiewicz theorem.

Run

```
python codes/foundations/a12_classii_source_square_reduction_verify.py
```

The expected integrated sentinel is `A12-CLASSII-SOURCE-SQUARE-REDUCTION-INTEGRATED-PASS`.

9 Devil's-advocate

1. **Objection:** a rational $B(S_{j-1}\phi)$ is not band-limited, so a polynomial six-linear support proof is invalid. **Verdict: DISMISSED.** The main proof uses the global Pauli/Fierz operator bound before any Fourier argument. Rational Fourier support is never truncated.

2. **Objection:** the previous Frobenius constant was silently replaced by a smaller guessed number. **Verdict: DISMISSED.** Equations (3.1)–(3.6) prove the operator constant, include the floor and third component, and identify a saturating tangent direction. Both executables reconstruct it.

3. **Objection:** the shell decay misses a factor two, three, or the physical $2\pi/L$. **Verdict: DISMISSED.** The shell begins at the integer $N_{j-1}+1$, D^* has symbol norm $|k|$, and all derivative directions are already in that norm. The executable enumerates the production symbol.

4. **Objection:** A9 requires M_R^4, so (6.2) is undercounted. **Verdict: DISMISSED.** A9's Schatten square has two G_j legs. Here ℓ_j has one G_j^* and its Hilbert norm is squared, giving M_R^2. The primary audit rejects the fourth-power negative control.

5. **Objection:** bounded individual sharp projections imply the maximal theorem. **Verdict: VALID WITH MITIGATION.** That implication alone is false. Section 5 instead proves a strict-dyadic smooth-plus-annular square-function estimate using product Marcinkiewicz variation.

6. **Objection:** sampled FFT ratios certify the infinite-cutoff constant. **Verdict: UPHELD.** They do not. This package records T4, leaves the numerical enclosure gate open, and does not call (7.2) proved.

7. **Objection:** the theorem extends unchanged to radial balls or to nearly continuous shell ratios. **Verdict: UPHELD.** The cube-face Marcinkiewicz structure and strict dyadic scale kernel are explicit hypotheses. No such extension is claimed.

10 References

1. L. Grafakos and L. Slavíková, "The Marcinkiewicz multiplier theorem revisited," arXiv:1706.07201.

2. M. Mastyło and L. Rodríguez-Piazza, "Convergence almost everywhere of multiple Fourier series over cubes," *Trans. Amer. Math. Soc.* **370** (2018), 1629–1659, doi:10.1090/tran/7172.

3. TECT claims A8–A11 and their hash-pinned manifests.

11 Result footer

```
Result ID:                A12-CLASSII-SOURCE-SQUARE-REDUCTION
Precise statement:        The A11 adapted source-square is bounded by
                          (beta_op^2/c_sym) M_R^2 M_6^4 Q_6^2
                          ||phi||_6^6, uniformly in the cutoff.
Scope:                    L=16 fixed-floor production coefficients;
                          six-real fields; scalar regulator;
                          strict sharp-cube filtration.
Dependencies:             A1-PRODUCTION-FUNCTIONAL-REALISATION;
                          A7--A11 Class-II chain (full IDs in card).
Evidence grade:           ANALYTIC, EXACT, EXECUTED.
Reproduction command:     python codes/foundations/
                          a12_classii_source_square_reduction_verify.py
Expected output:          A12-CLASSII-SOURCE-SQUARE-REDUCTION-
                          INTEGRATED-PASS; all assertions pass; exit 0.
Falsification gate:       Any failed Fierz identity, shell factor,
                          M_R power, theorem hypothesis, or hash.
Tier before / after:      n/a (new claim) / T4.
No-overclaim statement:   This result does not supply a certified decimal
                          enclosure, T-047 closure, sextic reserve,
                          Nelson, or T5/T6/T7.
Next required action:     Enclose H_6 and recompute the sextic budget.
```